

2026/09/15 (WEEK 2)

1. INTRODUCTION TO ERROR-CORRECTING CODES

MOTIVATION

There are four types of coding:

1. Data compression (source coding) e.g. ZIP data compression
2. Error control (channel coding): adds useful redundancy to the data from a source to make the transmission more robust to disturbances present on the transmission channel $\rightarrow$ our focus
3. Cryptographic coding: secure communication in the presence of third parties
4. Line coding

Examples where we need error-correcting codes: scratched disc, telecommunications, deep-space optical communication

$\star$ There is no way to guarantee accuracy for error-correcting codes!

DEFINITION

A repetition code is where something is sent multiple times in a row

Q-ARY CODES

DEFINITIONS

A q-ary code is a given set of sequences of symbols where each symbol is chosen from a set $F_q = \{\lambda_1, \lambda_2, \dots, \lambda_q\}$ of q distinct elements

$\hookrightarrow$ The set F_q is called the alphabet

We often take F_q to be $\mathbb{Z}_q = \{0, 1, 2, \dots, q-1\}$

Binary codes are 2-ary codes

Ternary codes are 3-ary codes

EXAMPLE

The set of all words in the English language is a code over the 26-letter alphabet $\{A, B, \dots, Z\}$

EXAMPLE

The set of all street names in the city of Salford is a 27-ary code (" " is the 27th symbol)

$\hookrightarrow$ It's an example of poor encoding $\rightarrow$ two streets are HILLFIELD DRIVE and MILLFIELD DRIVE

CODEWORDS

DEFINITION

A code where each codeword is a sequence consisting of a fixed number n of symbols is called a block code of length n

$\hookrightarrow$ For this course, "code" refers to "block code"

EXAMPLE

We want to encode 4 messages into binary codewords: N, S, E, W. How?

The fastest (shortest) code we can use is:

$$C_1 = \begin{Bmatrix} 00 \\ 01 \\ 10 \\ 11 \end{Bmatrix} = \begin{matrix} N \\ W \\ E \\ S \end{matrix} \Rightarrow \text{we can identify N, W, E, S with the four vectors of } (\mathbb{F}_2)^2$$

However, we can add an extra digit to detect any single error. We say it is a single-error-detecting code

$$C_2 = \begin{Bmatrix} 000 \\ 010 \\ 100 \\ 110 \end{Bmatrix}$$

If we add two more digits, we get a length 5 code that can correct any single error, since the received vector is closer to the transmitted codeword than to any other

HAMMING DISTANCE

DEFINITION

The (Hamming) distance between two vectors x and y of $(F_q)^n$ is the number of places in which they differ. It's denoted by $d(x, y)$. It is a legitimate distance function: (metric space)

(i) $d(x, y) = 0 \Leftrightarrow x = y$

(ii) $d(x, y) = d(y, x) \quad \forall x, y \in (F_q)^n$

(iii) $d(x, y) \leq d(x, z) + d(z, y) \quad \forall x, y, z \in (F_q)^n$

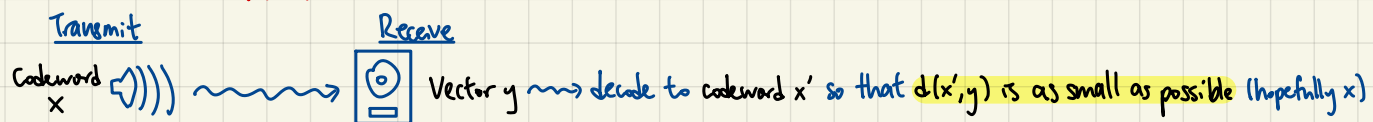
EXAMPLE

In $(F_2)^5$, $d(00111, 11001) = 4$

In $(F_3)^4$, $d(0122, 1220) = 3$

DECODING

SETTING (NEAREST NEIGHBOR DECODING)

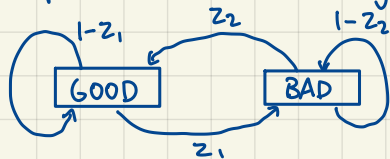


ASSUMPTIONS

- Each symbol transmitted has the same probability $p < 0.5$ of being received in error — p is called the symbol error probability
 - If a symbol is received in error, all of the $q-1$ possible errors are equally likely
- $\rightarrow$ Such a channel is called a q -ary symmetric channel

GILBERT-ELIOT CHANNEL MODEL

The probabilities the transmitted symbols are in GOOD or BAD (error) states is as follows:



In other words, errors come in a consecutive manner and the channel remains some time in that state

MINIMUM DISTANCE DECODING

DEFINITION (MINIMUM DISTANCE OF A CODE)

$$d(C) = \min \{d(x, y) \mid x, y \in C, x \neq y\}$$

DEFINITION OF AN (n, M, d) -CODE

An (n, M, d) -code is a code of length n , containing M codewords, and having minimum distance d

EXAMPLES

$C_1 = \begin{Bmatrix} 000 \\ 001 \\ 010 \\ 100 \end{Bmatrix} \Rightarrow d(C_1) = 1$; C_1 is a $(2, 4, 1)$ -code

$C_2 = \begin{Bmatrix} 000 \\ 001 \\ 010 \\ 100 \end{Bmatrix} \Rightarrow d(C_2) = 2$; C_2 is a $(3, 4, 2)$ -code

$C_3 = \begin{Bmatrix} 0000 \\ 0010 \\ 0100 \\ 1000 \end{Bmatrix} \Rightarrow d(C_3) = 3$; C_3 was a $(5, 4, 3)$ -code

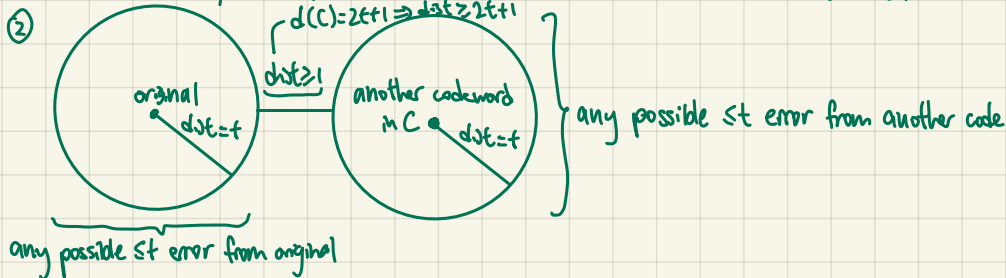
C = the q -ary repetition code of length n is an (n, q, n) -code

THEOREM (SUFFICIENT CONDITIONS FOR DETECTION AND CORRECTION)

- ① A code C can detect $\leq s$ errors in any codeword if $d(C) = s+1$
- ② A code C can correct $\leq t$ errors in any codeword if $d(C) = 2t+1$

Intuition

- ① If # errors $\leq s$, then the errored code doesn't exist in C (distance $\geq s+1$), so it can be corrected



Since the spheres don't touch, we can tell what the original code is with certainty for $\leq t$ errors

COROLLARY

minimum distance d

If a code C has $d(C) = d$, then C can be used either to

- ① detect $\leq d-1$ errors
- ② correct $\leq \lfloor \frac{d-1}{2} \rfloor$ errors in any codeword

2. THE MAIN CODING THEORY PROBLEM

A good (n, M, d) -code has:

- Small n (for fast transmission of messages)
- Large M (to enable transmission of a wide variety of messages)
- Large d (to correct many errors)

MAIN PROBLEM

(n)

(n)

(d)

What is the largest code of given length and minimum distance?

DEFINITION OF $A_q(n, d)$

$A_q(n, d)$ is the largest value of M s.t. $\exists$ a q -ary (n, M, d) -code

EXAMPLES

(i) $A_q(n, 1) = q^n$ (directly list all 0 or $q-1$ for each digit)

(ii) $A_q(n, n) = q$ (repetition code of length n)

DEFINITION OF EQUIVALENT CODES

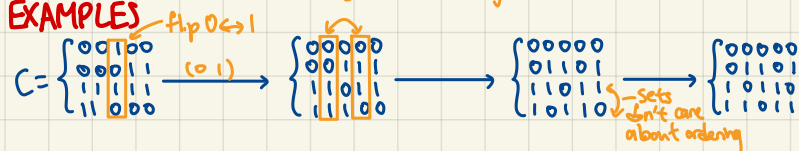
Two q -ary codes are called equivalent if one can be obtained from the other by a combination of operations of the following types:

1. Permutation of the positions of the code
 2. Permutation of the symbols appearing in a fixed position
- ★ Note: it's permutation (i.e. bijection)

If the code is displayed as an $M \times n$ matrix whose rows are the codewords, then these correspond to

1. Rearranging the columns of the matrix
 2. Relabelling the symbols appearing in a given column
- Needs to be systematic though

EXAMPLES



$$C = \left\{ \begin{bmatrix} 0 & 1 & 2 \\ 1 & 2 & 0 \\ 2 & 0 & 1 \end{bmatrix} \xrightarrow{(102)} \begin{bmatrix} 0 & 0 & 2 \\ 1 & 1 & 0 \\ 2 & 2 & 1 \end{bmatrix} \xrightarrow{(012)} \begin{bmatrix} 0 & 0 & 0 \\ 1 & 1 & 1 \\ 2 & 2 & 2 \end{bmatrix} \right.$$

LEMMA

Any q -ary (n, M, d) -code over an alphabet $\{0, 1, 2, \dots, q-1\}$ is equivalent to an (n, M, d) -code which contains the all-zero vector $0 = 00 \dots 0$ ($\Rightarrow$ wlog, we can assume 0 is in our q -ary (n, M, d) code)

Proof

Choose any codeword $x_1 x_2 \dots x_n$ and for each $x_i \neq 0$, apply the following permutation to the symbols in position i :

$$\begin{pmatrix} 0 & x_i & j \\ \downarrow & \downarrow & \downarrow \\ x_i & 0 & j \end{pmatrix} \quad \forall j \neq 0, x_i$$

Then, $x_1 x_2 \dots x_n \rightarrow 0 \dots 0$ in an equivalent code $\square$

OPERATIONS ON $(\mathbb{F}_2)^n$

DEFINITIONS OF SUM AND INTERSECTION

Let $x = x_1 \dots x_n, y = y_1 \dots y_n \in (\mathbb{F}_2)^n$, then:

- The sum $x + y = (x_1 + y_1, x_2 + y_2, \dots, x_n + y_n)$

- The intersection $x \cap y = (x_1 y_1, x_2 y_2, \dots, x_n y_n)$

where all calculations are modulo 2 without carrying

Ref these Cayley Tables:

+	0	1
0	0	1
1	1	0

·	0	1
0	0	0
1	0	1

$(x \cap y = x \cdot y)$

EXAMPLES

$$11100 + 00111 = 11011$$

$$11100 \cap 00111 = 00100$$

DEFINITION OF WEIGHT

The weight of a vector x in $(\mathbb{F}_2)^n$ is defined to be the #1's in x

LEMMA

If $x, y \in (\mathbb{F}_2)^n$, then $d(x, y) = w(x + y)$

LEMMA

If $x, y \in (\mathbb{F}_2)^n$, then $d(x, y) = w(x) + w(y) - 2w(x \cap y)$

THE RELATION BETWEEN ODD AND EVEN d

THEOREM

Suppose d : odd. Then, $\exists$ binary (n, M, d) -code $\Leftrightarrow \exists$ binary $(n+1, M, d+1)$ -code

Proof Sketch

" $\Rightarrow$ ": Add an overall parity check to the code C

$$x = x_1 \dots x_n \rightarrow \hat{x} = \begin{cases} x_1 \dots x_n 0, & w(x): \text{even} \\ x_1 \dots x_n 1, & w(x): \text{odd} \end{cases}$$

" $\Leftarrow$ ": Removing one position where digits differ $\square$

COROLLARY

If d : odd, then $A_2(n+1, d+1) = A_2(n, d)$. Equivalently, if d : even, then $A_2(n, d) = A_2(n-1, d-1)$

SPHERES IN $(F_q)^n$

DEFINITION

$\forall u \in (F_q)^n$, integer $r \geq 0$, the sphere of radius r and center u , denoted $S(u, r)$, is the set $\{v \in (F_q)^n \mid d(u, v) \leq r\}$
 $\hookrightarrow$ If $d(C) \geq 2t+1$, then $S(u, t)$ are disjoint $\forall u$: codeword of C (i.e. no overlap)

OBSERVATION

A sphere of radius r in $(F_q)^n$ ($0 \leq r \leq n$) contains exactly $\binom{n}{0} + \binom{n}{1}(q-1) + \binom{n}{2}(q-1)^2 + \dots + \binom{n}{n}(q-1)^n$ vectors
 $\hookrightarrow \binom{n}{k} \leadsto$ choose which k bits to make wrong
 $\hookrightarrow (q-1)^k \leadsto$ # ways to make those k bits wrong

DEFINITION (SPHERE-PACKING/HAMMING BOUND)

The sphere-packing/Hamming bound is:

A q -ary $(n, M, 2t+1)$ -code satisfies $M \underbrace{\left\{ \binom{n}{0} + \binom{n}{1}(q-1) + \dots + \binom{n}{t}(q-1)^t \right\}}_{\text{\# spheres}} \leq q^n$ enough to have distinct codes of all fuzzy M codes so they don't overlap ($q^n =$ # length n q -ary codewords)

PERFECT CODES

DEFINITION

Perfect codes are codes which achieve the Hamming bound

$\hookrightarrow$ Perfect t -error-correcting codes have M spheres of radius t centered on codewords that fill the space $(F_q)^n$ w/o overlapping (i.e. every vector in $(F_q)^n$ is at distance $\leq t$ from exactly one codeword)

EXAMPLES OF TRIVIAL PERFECT CODES

1. The whole $(F_q)^n$
2. The binary repetition code of length n , n : odd $\leadsto \{0 \dots 0, 1 \dots 1\}$